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Lightweight Object Detection for African Penguin Population Monitoring

BACKGROUND: Evolution of Object Detection

Advancements in computer vision and machine learning have transformed penguin conservation research:

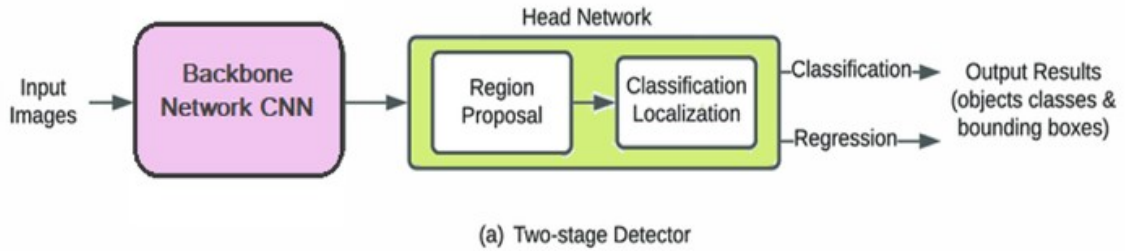
<u>TWO-STAGE DETECTORS:</u>
Detection → 2 separate stages:

1. Region Proposal Stage: Generate candidate object regions (possible bounding boxes)

2. Classification & Refinement Stage: Classify regions and refine its bounding box

Examples: Faster R-CNN *(outdated technology)*
Evolution: R-CNN → Fast R-CNN → Faster R-CNN

Characteristics:
- High detection accuracy (historically)
- Strong performance on small or complex objects
- Computationally intensive
- Slower inference
- **Traditionally GPU-dependent**



(a) Two-stage Detector

Mohammed, S. Y. (2025). Architecture review: Two-stage and one-stage object detection. *Franklin Open*, 100322.

BACKGROUND: Evolution of Object Detection

Advancements in computer vision and machine learning have transformed penguin conservation research:

SINGLE-STAGE DETECTORS

Detection → 1 unified regression step

- Image is processed once
- Bounding boxes and class probabilities are predicted at the same time

Characteristics:

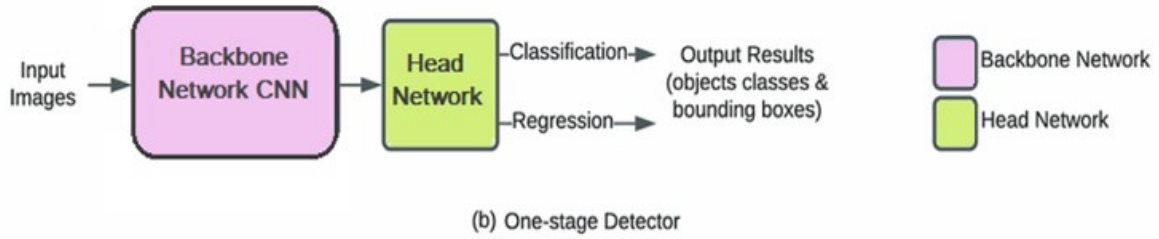
- End-to-end regression
- Faster inference
- More computationally efficient
- Suitable for real-time applications
- **Better aligned with CPU or edge deployment**

Examples:

YOLO (You Only Look Once)

YOLOv5n, YOLOv8n

MobileNet-based detectors



(b) One-stage Detector

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African Penguin Detection Studies & Deployment Gap

- **Aerial Imagery Detection**
- *Wu et al. – YOLOPd (Single-Stage)*
- Designed address small-object detection
- +8.5 mAP over Faster R-CNN
- 94.6% counting accuracy
- Evaluated on V100 GPU
- **No CPU deployment analysis**



African Penguin Detection Studies & Deployment Gap

- **<u>Aerial Imagery Detection</u>**
- *Liu et al. – Faster R-CNN (Two-Stage)*
- UAV orthomosaic imagery
- Result within 2% of expert counts
- Required GTX-1070 / GTX-2080 GPUs
- **No CPU efficiency evaluation**





African Penguin Detection Studies & Deployment Gap

- **Pose-Based Alternative**
- *Van den Berg – APE Model*
- Keypoint regression instead of bounding boxes
- 40x faster CPU pipeline than state-of-the-art models
- Effective for behaviour analysis
- **Pose complexity doesn’t align with simple counting methods**

African Penguin Detection Studies & Deployment Gap

•**Hardware-Aware Optimization**

- Hardware-specific optimization for Intel CPUs/ iGPUs
- 126x speedup reported *(Andriyanov, 2020)*
- +1.5x improvement using integrated GPU *(Zunin, 2021)*

GAP:

•Strong detection accuracy under GPU-based conditions.

•**CPU-only or resource-constrained deployment = unexplored**



Problem Statement

African Penguin *(Spheniscus demersus)*

- Endemic species to coastline of Namibia and South Africa
- +/- 73% population decline → IUCN Red List of Threatened/ Endangered Species

Problem:

•Field surveys typically conducted on laptop systems without dedicated GPUs and limited connectivity

•Makes real-time deployment of high-capacity models impractical

•Trade-off between efficiency gains and census-level accuracy remains insufficiently explored under CPU-only, hardware-constrained field conditions.



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Research Questions

RQ1: What level of detection accuracy can be achieved using lightweight object detection models for African Penguin monitoring in multi-instance, real-world conditions?

RQ2: What accuracy–efficiency trade-offs emerge when comparing lightweight and higher-capacity variants of the same detection architecture under CPU-constrained conditions?

RQ3: How does hardware-aware optimization influence inference latency and computational efficiency under CPU and integrated GPU deployment?

Research Objectives

- RO1: Prepare datasets and develop a lightweight object detection model to measure detection accuracy in real-world conditions.

- RO2: Evaluate inference latency and computational efficiency under CPU and integrated GPU deployment.

- RO3: Analyse accuracy–efficiency trade-offs between lightweight and higher-capacity models under CPU constraints.





Approach

1. Dataset preparation (AMP, SSP, Open Images subset)
2. Automated bounding box generation from existing pose keypoints
3. Training object detection models (possibly YOLOv8n / YOLOv8m) on GPU hardware
4. Optimising trained models for CPU inference testing using Intel’s OpenVINO toolkit
5. Accuracy–efficiency analysis

Why is this Computer Science?

1. Research Gap: Existing penguin detection methods rely on GPU-heavy, high-resolution environment, CPU-only resource-constrained deployment hasn't been addressed to solve conservation efforts

2. Comparison of lightweight and higher-capacity YOLO variants (direction for future exploration: NMS vs NMS-free)

3. Quantifying accuracy-efficiency trade-offs with metrics, mAP, F1, mAE etc.

4. Explores systems optimization: Using inference engines (OpenVINO) and hardware CPU, iGPU, GPUs

5. Contributes to ecological monitoring and CS on efficient ML deployment representative of field deployment



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Timeline

Date	Milestone / Activity
23 Feb 2026	Draft proposal submission
24 Feb – 10 Mar 2026	Seminar Series 1: Proposal presentation and feedback integration
16 Mar 2026	Finalised proposal submission
4 May 2026	Literature Review submission (updated problem statement, questions, objectives)
May 2026	Dataset preparation and bounding box generation script implementation
15 Jun 2026	Begin drafting research design / methodology chapter
VAC (Jun/Jul)	Model implementation and baseline training
7 - 14 Jul 2026	Seminar Series 2: Project implementation presentation
31 Jul 2026	Progress report submission
Aug 2026	OpenVINO optimization and CPU/iGPU benchmarking
Sep 2026	Accuracy–efficiency trade-off analysis and statistical evaluation
18 Sep 2026	Full draft research report submission
2 Oct 2026	Short paper submission (if required)
12–14 Oct 2026	Seminar Series 3: Final results presentation
16 Oct 2026	Final thesis submission
5 Nov 2026	Oral examination
6 Nov 2026	Final corrected submissions (report and ACM short paper)

References

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Q & A